

GIRISH ARUN KOUSHIK

Guildford, United Kingdom

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SUMMARY

PhD Researcher and Chess Grandmaster specializing in detection and moderation of Multimodal Hate Speech and other online harmful content. Additionally, spent a great deal of time applying Reinforcement Learning to Large Language Models (LLM-RL) for different applications. I apply the same strategic rigor to building robust models as I do to calculating complex chess positions. I am seeking research roles to tackle difficult multimodal applications and challenges.

EDUCATION

PhD in Computer Science

University of Surrey

2023 – Present

Guildford, United Kingdom

- Multimodal framework for Hate Speech Detection on Social Media Platforms - supervised by Dr. Diptesh Kanojia and Prof. Helen Treharne.

MSc in Advanced Computer Science with Artificial Intelligence

University of Strathclyde

2022 – 2023

Glasgow, United Kingdom

BE in Electronics and Communication Engineering

R.V. College of Engineering (Visvesvaraya Technological University)

2015 – 2019

Bengaluru, India

PAPER PUBLICATIONS

1. **Koushik, G. A.**, Treharne, H., & Kanojia, D. (Jan 2026). *TANDEM: Temporal-Aware Neural Detection for Multimodal Hate Speech*. Accepted to AACL-ICWSM 2027.
2. Sindhuja A., **Koushik, G. A.**, Qian, S., Kanojia, D., & Orasan, O. (Dec 2025). *Beyond Scalar Scores: Reinforcement Learning for Error-Aware Quality Estimation of Machine Translation*. Accepted to the Cambridge NLP Journal.
3. **Koushik, G. A.**, Fatemeh, N., Birch, K., Qian, S., & Kanojia, D. (Aug 2025). *The Mind's Eye: A Multi-Faceted Reward Framework for Guiding Visual Metaphor Generation*. Accepted to GenProCC Workshop at NeurIPS 2025.
4. **Koushik, G. A.**, Treharne, H., Joshi, A., & Kanojia, D. (April 2025). *TRACE: Textual Relevance Augmentation and Contextual Encoding for Multimodal Hate Detection*. Accepted to the AISI Track at AACL 2026.
5. Saeid, A., Sabu, A., **Koushik, G. A.**, Neri, F., & Kanojia, D. (Aug 2025). *Cyberbullying Detection via Aggression-Enhanced Prompting*. Accepted to RANLP 2025.
6. Feng, X., Liu, B., Song, Y., Fu, H., Wan, Z., **Koushik, G. A.**, Hu, Z., Yang, M., Wen, Y., & Wang, J. (Nov 2024). *Natural Language Reinforcement Learning*. Accepted to SSI-FM Workshop at ICLR 2025.
7. **Koushik, G. A.**, Kanojia, D., & Treharne, H. (Dec 2024). *Towards a Robust Multimodal Framework for Hate Detection: A Study on Video vs. Image-based Content*. Accepted to the MM4SG Workshop at WWW'2025.

WORK EXPERIENCE

Research Assistant in Multimodal Retrieval

eBay US (Remote)

Nov 2025 – Present

Guildford, United Kingdom

Teaching Assistant in 'Natural Language Processing'

University of Surrey

Feb 2025 – May 2025

Guildford, United Kingdom

- Laboratory demonstrator for postgraduates in COM3029/COMM061 Natural Language Processing, taught by Dr. Diptesh Kanojia and Dr. Suparna De.
- Topics covered: Tokenization, Feature Vectors & Embeddings, Transformers, Large Language Models, Instruction Fine-tuning, and Retrieval-Augmented Generation.

Teaching Assistant in 'Foundations of AI'

University of Strathclyde

Jan 2023 – Mar 2023
Glasgow, United Kingdom

- Laboratory demonstrator for Year 3 undergraduates in CS310 Foundations of AI, taught by Dr. Andrew Abel.
- Topics covered: Search Trees, Game Trees, Logic & Planning, and Neural Networks.

Software Development Engineer (AI/ML)

Diatoz Solutions Private Limited

Jan 2021 – Aug 2022
Bengaluru, India

- Development: AI/ML and NLP models.
 - Developed and deployed models for 'e2eHiring', a candidate recruitment platform.
 - Added features: smart hire, candidate and job recommendation system, assessment module, resume parsing service, online video proctoring model.
 - ML open-source contribution: added a feature for haystack (*deepset.ai*).
- Mentorship: trained two interns on machine learning and deep learning.
- Integration: facilitated cross-team collaboration between the AI/ML team and the back-end team.

TECHNICAL SKILLS

Programming Languages and Databases	Python, Latex, R, SQLite, MongoDB
Applications and Services	Git, AWS, Google Cloud, Docker
Frameworks and Tools	PyTorch, TensorFlow, vLLM, Unsloth
Operating Systems	Windows, Linux, MacOS

PROJECTS

Multistream Audiovisual Speech Recognition

MSc project at University of Strathclyde

- Investigated compact Gabor filter features for visual speech recognition on a challenging large-vocabulary continuous speech dataset (LRS2). Developed and evaluated CNN, RNN, and transformer models using Gabor features for viseme classification and sequence-to-sequence prediction.
- Analysis provided insights into trade-offs between engineered intuitive features, such as Gabor filters, versus learnt representations for audiovisual speech recognition.

Recruitment Technology Platform (e2eHiring)

Diatoz Solutions

- Developed comprehensive AI/ML solutions for end-to-end recruitment automation, including:
 - NLP models for the assessment of technical skills from interview videos (accuracy 80%, inference time ~20s).
 - Resume parsing API service processing 50 resumes in <120s with detailed information extraction.
- Created custom datasets and implemented models using TensorFlow, OpenCV, and NLP techniques.

AWARDS AND ACHIEVEMENTS

- Received the best paper award at the MM4SG workshop at WWW'25.
- Member of the "StrathClust" Team that participated in the CIUK 2023 Cluster Challenge, representing the University of Strathclyde and secured third place.
- Received 'Pat-On-Back' Award for excellent project performance, Diatoz Solutions (May 2021, July 2022).
- 63rd Chess Grandmaster from India, conferred in 2019 by the International Chess Federation (FIDE); also won several international medals, like the Silver Medal in the World Youth Chess Olympiad and Gold Medal in the Asian and National Youth Championships.